

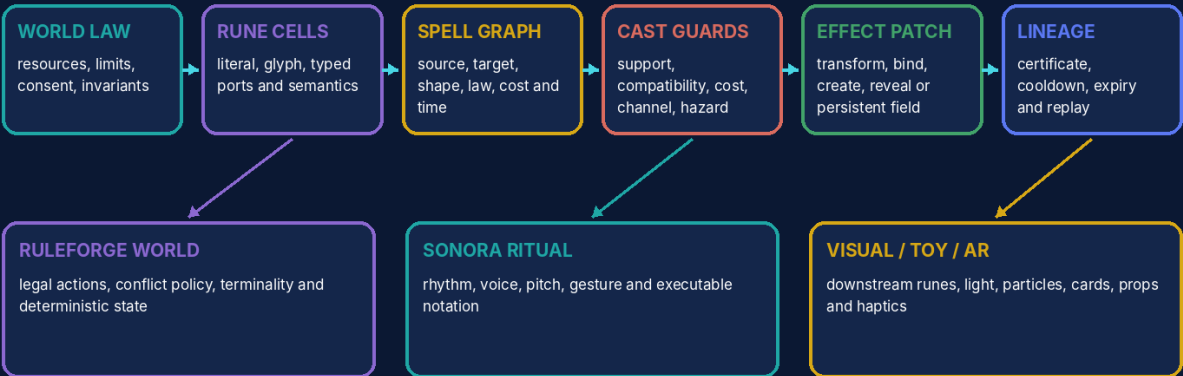
UNIVERSAL GEOMETRIC TOPOLOGICAL
OPEN MODULAR SUBSTRATE

UGTOMS ARCANA 1.0

Executable Magic Systems, Rune OperatorCells, Rituals and
Counterspell Cartridge

UGTOMS ARCANA 1.0 Architecture

A magic system becomes an executable rule language, not a bag of visual effects.



The cool part is a universal magic-system compiler whose visible runes cannot lie about the underlying rules.

Pure domain evaluation may propose evidence; only the retained authority chain may commit state.

ugtoms.domain.arcana@1.0.0

Component	ARCANA
Canonical identity	ugtoms.domain.arcana@1.0.0
Component type	Fictional domain cartridge
Version / date	1.0.0 / 30 August 2026
Document type	Open modular cartridge specification and implementation-ready profile
Status	Implementation-ready rules with deterministic fixtures; no supernatural or live-effects safety capability claimed

ATTRIBUTION AND EVIDENCE BOUNDARY

Prepared for Tom Klootwijk. Requester attribution is recorded as supplied and is not independently verified. This is a technical design artifact, not legal proof, scientific validation, regulatory approval or production safety certification.

Contents and Reading Rule

Section	Purpose
1	Executive decision and product fantasy
2	Source basis and evidence discipline
3	Open modular cartridge ABI
4	World-law profile and formal arcane state
5	Rune OperatorCells and executable notation
6	Spell graphs, ports and composition order
7	Support, targeting, resources and cast schedules
8	Effects, persistent fields, expiry and invariants
9	Counterspells, converse, inverses and conflict order
10	Ritual grammar, chance, cooperation and consent
11	Verified cast and atomic fictional-world patch
12	Gatefold spell rejection-repair-commit demonstration
13	Packing, projections and runtime adapters
14	Reference validation, boundaries and kill criteria
15	Complete 48-mechanism catalog
16	Claims ledger
17	High-impact profiles and final definition
READING RULE	
The UGTOMS source line supplies the reusable authority, typing, order, evidence and packing discipline. The specialist domain model in this cartridge is an engineering-derived extension. Where the source corpus is silent, this report says so instead of presenting general domain knowledge as source-derived.	

Executive Decision

DECISION - GO

Replace the previously planned generic EVIDENCE cartridge with ARCANA. ARCANA makes the full substrate legible and spectacular: runes are typed OperatorCells, spells are dependency graphs, counterspells are declared converse/counter relations, rituals are bounded grammars, and only a verified cast commits a fictional-world patch.

ARCANA is intentionally fictional. That makes it safe to go hard on the fantasy while remaining exact about the engine. A glowing rune can look wild, sing through SONORA, animate through a graphics runtime and live on a physical tile - but the rune cannot secretly change its type, cost, target, order or authority.

Dope AF capability	ARCANA contract
Universal magic-system compiler	A world profile declares which spell families, resources, limits, conservation rules and counters exist.
Runes that actually mean something	Literal symbol, glyph geometry, typed ports, semantics, converse relation and content hash share one address without collapsing.
Composable spells	Source, target, support shape, law, cost, timing and effects form a graph with explicit order and failure reasons.
Counterspell warfare	Inverse, converse, cancel, absorb, redirect, sever and residual effects are separate declared relations.
Ritual and co-op casting	Multiple casters, voices, gestures, instruments, anchors and schedules coordinate through deterministic phases.
Rule-safe spectacle	Particles, sound, haptics, AR and physical props remain projections of committed fictional state.

Source Basis and Evidence Discipline

The cartridge inherits a common UGTOMS rule: domain knowledge may be broad, but every authoritative change stays typed, ordered, evidence-bounded and replayable.

- Definitions, operators and profiles live inside the substrate with stable IDs, explicit dependencies and content hashes [S1-S3].
- Pure evaluation and model output remain upstream of support, compatibility, guards, verified proposals, deterministic commit and lineage [S1, S2, S4-S7].
- Coordinates, glyphs, pictures, meshes, simulations, scans and packed keys are not identity or authority by themselves [S1, S4-S10].
- Exactness, uncertainty, capability, budgets and error margins are explicit; UNKNOWN and REJECTED are valid outcomes [S1, S2, S5, S8-S10].
- Component-scoped identities keep this cartridge from masquerading as the newest universal version of every UGTOMS track [S11].
- Existing specialist cartridges remain peers and adapters; this report does not silently absorb their schemas or claims [S13-S17].

ENGINEERING-DERIVED DOMAIN CONTENT

The spell, rune, resource, target, effect, counterspell and ritual records are original fictional-domain engineering. The substrate sources contribute OperatorCells, explicit order, topology, guards, bounded grammar, deterministic game state, replay and projection boundaries. No supernatural or physical-effect claim is imported.

ID	Registered source	Use in this cartridge	Pages
S1	UGTS__FOUNDATION__UNICODE_v5.pdf	Foundation 5.0 OperatorCells: literal, glyph, semantic, field and converse faces remain co-addressed but typed separately.	pp. 1-21
S2	UGTS_KC_4_2_GeneralOperator_Order_Addendum(1).pdf	Typed OperatorSpec, dependency/phase/event/replay order, exactness, capability, budgets and adapter fallback policy.	pp. 2-32
S3	UGTS_KC_3_6_Tom_Klootwijk.pdf	Definitions as content-addressed substrate data, resolvable references, dependency closure and explicit operation order.	pp. 1-15
S4	Unified_Geometric_Topological_Substrate_GPU_Native_Addendum.pdf	Canonical support → compatibility → guard → verified event → transition → lineage authority and packed-state boundaries.	pp. 2-15
S5	UGTS_KC_2_0_Tom_Klootwijk_Expanded_Substrate.pdf	Patterns, fields, topology, SE(2)/SE(3), bounded dynamics, hybrid events and numerical/error contracts.	pp. 3-15
S6	UGTS_KC_Two_Hands_3_0_Interactive_Graphics_Runtime.pdf	Stable scene identity, proposals, deterministic commit, checkpoints, rollback, replay and divergence detection.	pp. 3-19
S7	UGTS_KC_3_9_KC_Elizabeth_Vector_Game_Runtime.pdf	Versioned projects, deterministic fixed-step runtime, snapshots/hashes and self-contained browser delivery.	pp. 2-16
S8	UGTS_KC_3_6_2_SCLP_Tom_Klootwijk.pdf	Time/phase/winding, finite topology maps, interval guards, finite grammar and exact packed-key boundaries.	pp. 1-15
S9	UGTS_KC_4_0_Spatial_Evidence_Ledger(1).pdf	Spatial observation versus authority, stable nodes/edges, uncertainty, route topology, repeat-scan change and replay.	pp. 3-15
S10	UGTS_KC_4_1_1_ATLAS_SPATIAL_SEED_SLAM_POCO_X7_PRO_MERGE_REPORT(1).pdf	Camera/IMU proposal path, bounded SLAM front end, stable IDs, ordered verifier and KSEED seed-plus-evidence boundary.	pp. 2-14
S11	UGTS_Versioning_Charter_Phase_1_Foundation_Chronology(1).pdf	Component-scoped identities and explicit release graphs instead of one misleading global ladder.	pp. 1-6
S12	UGTS_KC_Definition_Breaker_DB_0_1_Concept(1).pdf	Rules as physical objects, safe preview, hinges, winding, replay, seeded grammar and reason-coded commit.	pp. 2-24
S13	04_UGTOMS_RULEFORGE_1_0.pdf	Reusable rule-world host: typed actions, visibility, chance, conflict resolution, terminality and deterministic replay.	pp. 2-18
S14	06_UGTOMS_PROOFWEAVE_1_0.pdf	Proof graphs, small kernel, scoped certificates, trust boundaries and proposal-only search.	pp. 2-18

ID	Registered source	Use in this cartridge	Pages
S15	03_UGTOMS_FORGE_1_0.pdf	Parts, interfaces, datum frames, assembly order, inspection and infrastructure/fabrication boundaries.	pp. 2-16
S16	05_UGTOMS_VOLT_1_0.pdf	Typed networks, ports, capacities, events, faults, model adapters and physical-evidence separation.	pp. 2-17
S17	02_UGTOMS_SONORA_1_0.pdf	Executable notation, exact rational timing, cyclic structures, voices and downstream audiovisual projections.	pp. 2-15

Open Modular Cartridge ABI

The cartridge is a concrete instance of the shared domain object D_ARCANA:

D = (T, O, R, U, I, Q, E, G, Delta, L, Cert, K, Pi)	
Field	Contract
T	Entity, value and state types
O	Typed operators and transformations
R	Relations, graphs, topology and converse pairs
U	Units, dimensions, coordinate and profile conventions
I	Invariants, conservation laws and constraints
Q	Dependency, precedence, phase and reduction order
E	Observations, evidence, uncertainty and provenance
G	Guards, thresholds and admission predicates
Delta	Proposed and committed state patches
L	Lineage, replay, history and migration
Cert	Exactness, residual, interval, scope and status certificates
K	Cold atlas, hot codebook and packed representation
Pi	Optional visual, audio, physical or device projection

A conforming cartridge contains a manifest, type registry, operator registry, relation atlas, unit/profile system, dependency DAG, guard library, transition definitions, claims ledger, examples and validation fixtures. External mature libraries may sit behind adapters when they are safer or more accurate than a fresh reimplementation.

ARCANA cartridge package

Package record	Required role
world_law.json	Available magic families, resources, limits, invariants, authority mode and non-claims.
rune_atlas.json	Content-addressed literal/glyph/semantic cells, typed ports, aliases and converse/counter mappings.
spell_registry.json	Spell graphs, effects, costs, schedules, supports, targets and certificates.
ritual_grammars.json	Bounded productions, participants, phases, synchronization and failure/repair states.
conflict_profiles.json	Same-time priority, cancellation, redirection, absorption, residual and merge rules.
arcane_ledger.jsonl	Cast proposals, rejections, committed effects, persistent updates, expiry and lineage.
projections/	SONORA score, visual/AR, haptic, printed rune and physical-toy profiles.
fixtures/	Deterministic casts, invalid graphs, counterspell conflicts and replay checks.

General Operator Record and Order

```
OperatorSpec = (id, version, syntax, arity, domains, codomain, units, shape,
               laws, exactness, effects, capabilities, dependencies,
               order_contract, invariants, provenance, content_hash)
```

The record follows the General Operator and Order layer [S2]. A symbol or glyph is not allowed to smuggle in undeclared type, unit, order, inverse or authority semantics.

EXAMPLE RECORD

Operator ARC031 counterspell_relation
domain: SpellProposal x ActiveEffect x CounterProfile → CounterResult
relations: cancel | weaken | redirect | absorb | sever | no-match
order: support → compatibility → priority → cost → residual → patch
exactness: exact discrete inside one world-law profile
effects: proposal only until conflict resolver commits

Normative phase schedule

Phase	Meaning
1-3	parse, normalize and resolve content-addressed references
4-6	type, shape and unit validation; dependency closure
7-9	construct candidates; apply declared composition and reduction order
10-12	evaluate exact or bounded results; issue capability and residual certificate
13	support, compatibility and guard classification
14	verified proposal and deterministic conflict resolution
15	atomic commit, lineage, replay and optional projection

World-Law Profile and Formal Arcane State

A = (W, R, S, G, T, C, E, P, Delta, L, Cert)		
W fictional world-law profile and invariants		
R rune OperatorCell atlas and spell registry		
S casters, artifacts, anchors, targets and active fields		
G spell graphs, ritual grammars and dependency order		
T clocks, cast/travel/channel schedules and cooldowns		
C resource pools, costs, chance and conflict profiles		
E accepted/rejected cast and persistent-update events		
P downstream sound, visual, tactile and physical projections		
Delta atomic fictional-world effect patches		
L checkpoints, expiry, compensation and replay lineage		
Cert spell, counterspell, ritual and replay certificates		
World-law field	Examples	Why it must be explicit
Resource law	mana, charge, heat, memory, favor, cards, stamina, cooldown	No universal mana arithmetic or conservation rule is assumed.
Entity law	who/what may cast, target, resist, own, consent or maintain	A player-facing icon cannot define authorization.
Spatial law	range, line, volume, anchor, portal, ward, sheet or layer	Topology and support are rules, not visual magic.
Temporal law	instant, channel, travel, persistent, periodic, expiry, cooldown	Order and time affect results and must enter identity.
Effect law	transform, create, bind, reveal, project, summon, dispel	Every effect has domain, codomain, invariants and residual state.
Conflict law	priority, counter, absorption, reflection, merge, simultaneous resolution	No hidden engine rule decides which spell wins.
FICTIONAL AUTHORITY		
ARCANA may be exact inside a declared game or story world while making zero claim that the same symbols, laws or effects exist physically.		

Rune OperatorCells and Executable Notation

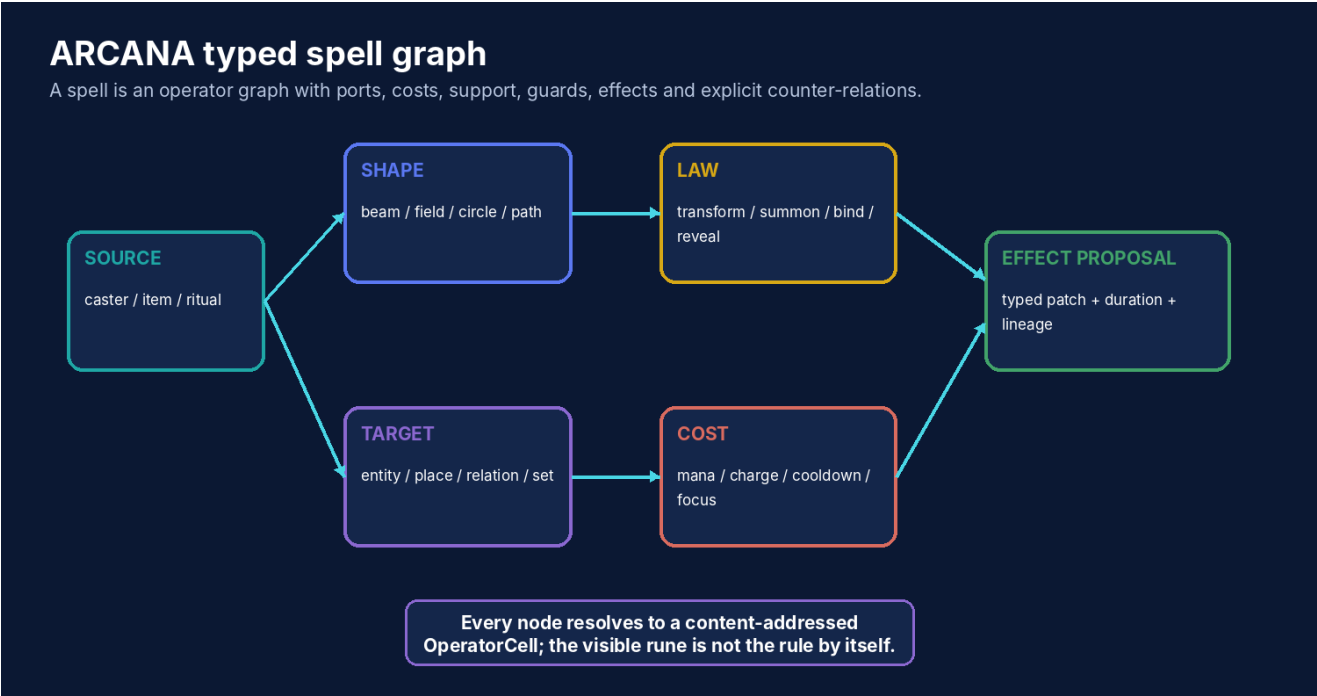


Figure 1. A spell is a typed operator graph; its visible rune is one face of a content-addressed definition.

Rune face	Role	Non-negotiable boundary
Literal	Exact Unicode sequence, token or fallback spelling.	A local opcode or font outline cannot replace global identity.
Glyph	Canonical rune geometry, optional SDF, tactile mark or physical outline.	Shape is presentation and interaction support, not semantic truth.
Typed ports	Source, target, resource, time, effect and trace connections.	Port direction/order cannot be guessed from left/right appearance.
Semantic operator	World-law transformation or relation.	Meaning is resolved from the OperatorCell and world profile.
Converse/counter map	Declared operand swap, inverse, cancellation or paired spell.	Mirroring alone does not reverse a spell.
Content hash/version	Exact definition identity and migration.	Hash integrity is not narrative truth, ownership or physical power.
RUNE COMPILER RULE		
The same rune cell feeds parsing, glyph generation, socket checking, codebook packing, visual/tactile projection, spell evaluation, debugging and tests. Prose and art cannot override the machine-readable record.		

Spell Graphs, Ports and Composition Order

```
SpellGraph = (source, focus, support_shape, target_query,
              law_nodes, resource_nodes, schedule_nodes,
              guards, effects, conflict_profile, trace_profile)

Example: Gatefold =
  resolve(caster, anchor_A, anchor_B)
  -> build_link(A,B, layer_profile)
  -> reserve(cost, cooldown)
  -> verify(target_clear, ward_policy, invariant)
  -> commit(portal_binding, duration, lineage)
```

Composition concern	Required contract
Dependency order	Every node lists stable dependencies; missing references and undeclared cycles reject.
Phase order	Resolve, shape, target, cost, channel, conflict, effect, persistent update and projection are distinct phases.
Noncommutativity	Amplify then split may differ from split then amplify; the graph records exact order.
Units/resources	Costs and outputs use world-specific typed quantities; addition and comparison require compatible profiles.
Partial/multivalued operators	Summon, reveal, random target or negotiation may return candidates or UNKNOWN rather than one hidden choice.
Effects vs presentation	An effect patch changes fictional state; particles, audio and animation are downstream.

NO FREE TEXT MAGIC

A production editor may offer natural language, drawing or gesture input, but it must compile to explicit spell nodes, profiles and guards before a cast can become authoritative.

Support, Targeting, Resources and Cast Schedules

Subsystem	Typed data	Typical rejection reason
Support	range, line/field/path, anchor, layer, line-of-effect, domain and event time	out-of-support, unresolved-anchor, wrong-layer
Targeting	entity/place/relation/set query, filters, count, order, consent and unknown policy	no-target, ambiguous-target, target-immune, consent-required
Resources	resource type, pool, reserve, cost vector, conversion and overdraw policy	insufficient-resource, incompatible-resource, reserved-by-other-cast
Channel/concentration	start, progress, interruption, concurrent limits and recovery	channel-broken, concentration-conflict, caster-unavailable
Travel/schedule	cast time, launch, arrival, duration, periodic tick, expiry and cooldown	too-late, cooldown-active, target-changed-before-arrival
Authorization/hazard	role, consent, ward, forbidden region, target protection and game-safety profile	unauthorized, protected-target, invariant/hazard-guard
<pre>cast_ready = in_support and target_compatible and resource_cost <= available_reserve and schedule/channel guards pass and authorization/consent passes and invariant/hazard margin remains positive</pre>		
CONSENT PROFILE		
In a fictional game, consent can still be a first-class rule: cooperative spells, possession, binding, mind-affecting or player-to-player effects may require explicit opt-in, role or mode guards.		

Effects, Persistent Fields, Expiry and Invariants

Effect family	State patch	Persistent obligations
Transform	change typed property, form, location, orientation or relation	preserve forbidden invariants; record reversibility and residuals
Create/summon	instantiate a declared entity/effect from a template or grammar	identity, ownership, lifetime, capacity and cleanup
Bind/enchant	add persistent relation, modifier, ward or obligation	source, target, strength, expiry, counter relation and maintenance
Reveal/illusion	change information or projection state under a viewer/profile	does not silently alter hidden authoritative object unless the world law says so
Persistent field	create support volume with periodic update/events	finite tick schedule, budget, collision/conflict and expiry
Dispel/expire	remove, weaken or transition an active effect	residual state, compensation and lineage remain visible
Invariant class	Example	
World conservation	total charge, card count or story-token budget under one declared profile.	
Identity preservation	transforming appearance does not silently replace target identity.	
Ownership/consent	binding or transfer cannot bypass declared actor relations.	
Topology	portal endpoints, layers, orientation and route class remain valid.	
Capacity	summoned entities, active fields or concentration slots remain within finite limits.	
Narrative/game state	quest, terminal, uniqueness or boss-phase invariants remain consistent.	

Counterspells, Converse, Inverses and Conflict Order

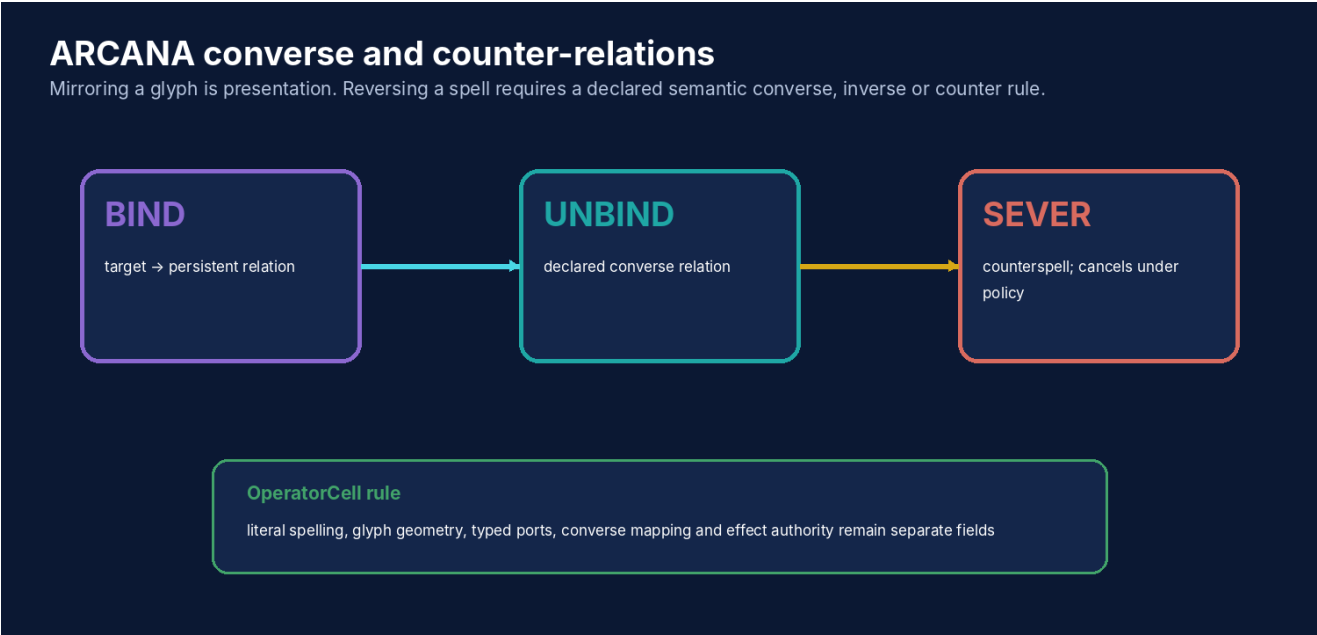


Figure 2. Bind, unbind and sever are distinct declared relations; mirroring the glyph does not choose among them.

Relation	Meaning	Example
Converse	Same relation viewed with canonical operand ports reversed.	A binds B ↔ B is bound by A under one declared relation.
Exact inverse	Applying the inverse restores the prior state inside its domain.	Temporary phase shift with unchanged external novelty.
Compensating spell	Creates a new patch that offsets an earlier effect.	Heal damage without erasing that the damage event occurred.
Counterspell	Cancels, weakens, redirects, absorbs or severs under a conflict profile.	Ward blocks only tagged projectile spells below a threshold.
Dispel/expiry	Transitions an active effect to inactive or residual state.	Enchantment expires but consumed resource remains spent.
Reflection	Redirects source/target or trajectory under explicit support and priority.	A mirrored rune alone is insufficient.
same_time_resolution = stable priority -> support overlap -> tag/target compatibility -> cost and authorization -> cancel / weaken / redirect / absorb / conflict -> one atomic batch patch + trace		
CONFLICT BOUNDARY		
Two visually colliding spells do not automatically interact. The world profile must declare compatible tags, supports, priorities and counter relations.		

Ritual Grammar, Chance, Cooperation and Consent

Ritual element	Contract
Participants	Caster roles, artifacts, anchors, voices, instruments and optional assistants with explicit authority.
Bounded grammar	Finite productions, maximum depth/symbols/branches, dependency closure and terminal status.
Synchronization	Beat, phase, gesture, location and event windows; exact rational timing may use SONORA.
Chance	Seeded finite distribution/profile, candidate set and lineage; randomness remains proposal input.
Repair	Reason-coded missing participant, wrong order, timing miss, resource conflict or unresolved rune.
Consent/cooperation	Each player/actor retains opt-in, cancellation, role and co-cast confirmation under the world profile.
RitualProduction: circle -> anchor_N + anchor_E + anchor_S + anchor_W chorus -> voice_A voice_B drum_phase gate -> circle ; chorus ; cost_lock ; destination_guard Budget: depth <= 4, symbols <= 64, participants <= 8, duration <= 90 s	
SEED BOUNDARY A ritual seed may reproduce declared timing, branch and presentation choices. It cannot reconstruct unstored player decisions, external input or committed novelty.	

Verified Cast and Atomic Fictional-World Patch

input / AI / player / ritual / item -> SpellProposal -> resolve world law + rune cells + dependencies -> support + target compatibility -> resource + schedule + authorization + invariant guards -> conflict / counterspell resolution -> verified cast -> atomic effect patch -> persistent schedules, cooldowns and expiry -> pre/post hashes + spell certificate + lineage -> sound, particles, animation, haptics, cards or AR projection	
Proposal field	Purpose
spell_id / version / world_profile	Fix exact rules and migration context.
caster / focus / anchors / targets	Stable identities and support context.
resource reservation	Prevents simultaneous overspend and records refund/consumption policy.
guard trace	Support, compatibility, schedule, authorization, consent, invariant and hazard reasons.
requested patch	Finite typed effect operations; no direct state writes.
conflict trace	Priority, counters, merge/cancel result and residuals.
pre/post hashes and lineage	Replay, divergence, expiry and compensation evidence.
SPECTACLE IS DOWNSTREAM	
The cast may fail even if an animation already previewed it. The world state may commit even if the screen effect is skipped. Presentation and authority remain separable.	

Gatefold Spell: Rejection, Repair and Commit

ARCANA Gatefold spell: rejection, repair, commit

The portal is fictional game semantics, but the type, order, guard and replay discipline is exact inside the declared world profile.

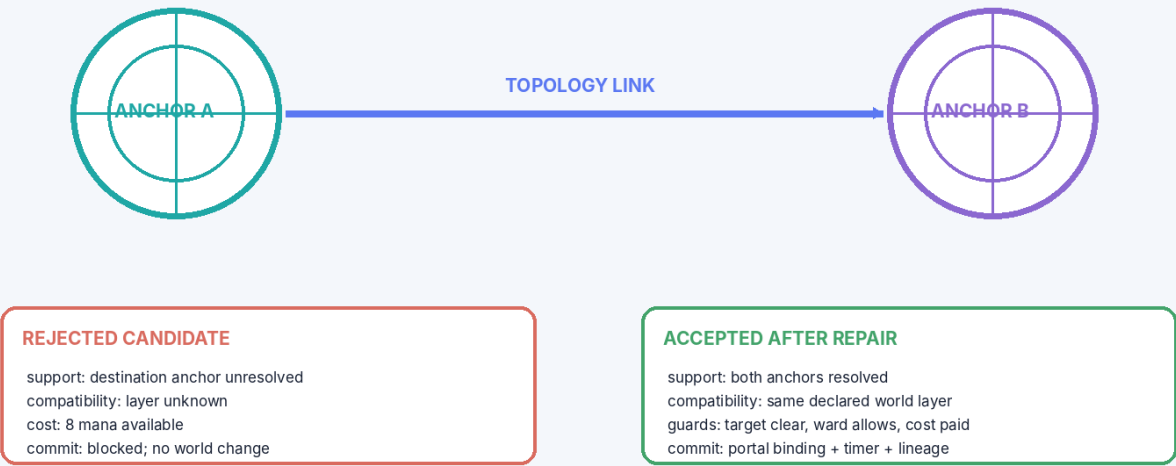


Figure 3. The same spell graph is rejected with an unresolved destination and accepted after typed repair.

Step	Candidate	Result
1	Resolve caster and Anchor A.	PASS: stable IDs, world layer and focus profile resolve.
2	Resolve Anchor B from a painted rune only.	REJECT: glyph exists but no destination entity/anchor identity is bound.
3	Player links rune to registered Anchor B.	PASS: literal and glyph now resolve to an exact anchor reference.
4	Build portal relation A → B.	PASS: same world layer, compatible topology and target-clear guard.
5	Reserve eight mana and cooldown slot.	PASS: pool has ten, no competing reservation.
6	Ward policy and invariant check.	PASS: permitted zone; no unique-portal or capacity violation.
7	Atomic commit.	Portal binding, expiry timer, resource delta, cooldown and lineage written together.
8	Replay.	The certificate reconstructs the same post-state; tampered target ID rejects.

DEMO BOUNDARY

The demonstration proves deterministic behavior inside the fictional profile only. It is not a claim about real portals, energy, symbols or supernatural effects.

Packing, Projections and Runtime Adapters

Layer	What may be compacted	What must stay explicit
Rune atlas	hot codebook IDs, shared glyph geometry and typed port schemas	cold global identity, version, converse/counter map and exactness
Spell graph	shared subgraphs, templates, parameter deltas and bounded grammar seeds	order, dependencies, costs, targets, guards and external novelty
Persistent effects	checkpoint plus deterministic update deltas	active identities, expiry, residuals, conflicts and lineage
Projection	procedural particles, SONORA motifs, animation and haptic profiles	authoritative effect state and state hash
Replay	initial checkpoint plus accepted cast/persistent events	rejected/accepted reason traces required by policy and any exogenous input
Adapter	Role and boundary	
RULEFORGE	Hosts legal actions, conflicts, terminality and world-state authority.	
SONORA	Compiles ritual timing, voices, pitch, rhythm and audio cues from the same spell/ritual IDs.	
Graphics / AR	Renders rune glyphs, fields, particles, portals and previews; never writes the spell ledger.	
Physical toys	Typed sockets, tactile rune faces, resource tokens, phase rail and replay beads.	
Narrative engine	Uses spell lineage, consequences and branch state without inventing rule changes.	
AI spell assistant	May propose graphs or explanations; every node must resolve and pass the verifier.	

Reference Validation, Boundaries and Kill Criteria

REFERENCE SELF-CHECK RESULT

32/32 deterministic specification self-checks passed. These checks cover catalog uniqueness, required fields, dependency closure, demo arithmetic and declared reason codes. They are package evidence, not independent domain validation.

Checked area	Result
Catalog integrity	48 unique ARC IDs; every mechanism has family, name and core contract.
Rune split	Literal, glyph, typed ports, semantics and converse/counter mappings remain separate fields.
Spell dependency fixture	Unresolved target anchor rejects; repaired graph closes without cycles.
Resource fixture	Reservation prevents double spend and applies cost/cooldown atomically with the effect.
Counterspell fixture	Mirroring alone has no effect; declared counter profile produces deterministic cancel/weaken/redirect status.
Persistent fixture	Expiry and periodic updates reproduce the same state hash under the declared schedule.
Replay fixture	Gatefold certificate reconstructs the post-state and rejects a tampered target.
Projection boundary	Sound, particles and rune geometry can be removed without changing authoritative spell state.

Explicit non-claims

- No supernatural, occult, physical-energy, healing, divination or real-world magic capability is claimed.
- No rune, word, gesture, sound, glyph geometry or mirror operation has authority outside a declared fictional profile.
- No pyrotechnic, electrical, chemical, medical, psychological or live-action safety approval is provided.
- No universal magic ontology, mana law, counterspell inverse, ritual grammar or balance result is claimed.
- No game engine, audio engine, graphics runtime or physical-toy implementation is completed by this PDF.
- No content hash, replay or spell certificate proves narrative ownership or truth outside the cartridge.

Kill criteria

1. Reject or narrow a use when a symbol, codebook slot or packed record cannot resolve to its exact global definition.
2. Reject when a capability or exactness label is stronger than the model, measurement or external source supports.
3. Reject when units, profiles, coordinate systems, operand order or dependency order are implicit.
4. Reject when a proposal bypasses support, compatibility, guards, deterministic commit or lineage.
5. Reject when an external adapter is required for safety or conformance but its version, calibration or provenance is absent.
6. Reject any performance, compression or safety claim without a named workload, target, error/equivalence contract and evidence.

Complete ARCANA 1.0 Mechanism Catalog

The catalog contains 48 cartridge-local mechanisms. These IDs are component-scoped and do not silently extend a global M-number ladder.

ID	Family	Mechanism	Core contract
ARC001	World	World-law profile	Defines the fictional magic system, available operator families, conservation rules, limits and authority mode.
ARC002	Identity	Arcane entity	Stable identity for caster, target, creature, place or construct independent of its visual effect.
ARC003	Identity	Caster profile	Typed capabilities, permissions, channels, resources and known spell definitions.
ARC004	Identity	Target profile	Entity/place/relation/set target with admissible effect classes and consent/ownership policy.
ARC005	Identity	Artifact and focus	Stable item identity with slots, charges, attunement, bindings, provenance and break/repair state.
ARC006	Topology	Anchor identity	Stable spatial/topological anchor for portals, wards, summons and persistent fields.
ARC007	Notation	Rune OperatorCell	Literal rune, canonical glyph, typed semantic operator, ports, converse rules and content hash.
ARC008	Notation	Rune atlas and codebook	Cold global atlas plus hash-bound local aliases; unknown runes reject instead of aliasing.
ARC009	Typing	Spell port	Source, target, shape, resource, condition, effect and trace ports with explicit domain/codomain.
ARC010	Definition	Spell graph	Content-addressed acyclic or declared cyclic operator graph with dependencies and phase order.
ARC011	Definition	Spell family	Parameterized operator family with capability, cost and world-profile constraints.
ARC012	Notation	Literal/glyph/semantic split	A mirrored, rotated or decorated rune does not change semantic authority unless the profile declares it.
ARC013	Support	Spell support volume	Point, path, cone, region, network, relation or topological support with bounded capability.
ARC014	Targeting	Target query	Finite candidate query plus visibility/knowledge policy; targeting display never commits an effect.
ARC015	Topology	Portal and link relation	Explicit source/destination anchors, orientation, direction, capacity and persistence.
ARC016	Topology	Ward and boundary relation	Typed boundary that admits, blocks, redirects or taxes declared spell/effect classes.
ARC017	Resource	Arcane resource type	Mana, charge, focus, heat, attention or fictional essence with units/range and source profile.
ARC018	Resource	Resource pool	Authoritative current/max/regeneration/reservation state with nonnegative commit invariant.
ARC019	Resource	Cost vector	Finite resource consumption, deposit, debt or sacrifice under the selected world-law profile.
ARC020	Time	Cooldown state	Next admissible cast time, charges and reset policy remain explicit state.
ARC021	Time	Concentration channel	Persistent ownership/focus relation with interruption, capacity and handover policy.
ARC022	Time	Cast and travel schedule	Wind-up, release, propagation, duration, tick and expiry phases with deterministic order.
ARC023	Relation	Source-of spell	Declares which caster, artifact, ritual or environment is the authoritative source.

ID	Family	Mechanism	Core contract
ARC024	Relation	Effect-on target	Typed relation between a verified spell event and affected entities/places/relations.
ARC025	Effect	Transform effect	Proposes a bounded change of typed attributes, state, frame or relation under invariants.
ARC026	Effect	Create or summon effect	Creates a new fictional entity/construct with stable identity, ownership, lifetime and resource source.
ARC027	Effect	Bind or enchant effect	Creates a persistent relation or modifier with duration, stacking, removal and lineage.
ARC028	Effect	Reveal and illusion projection	Changes knowledge or presentation under a declared observer profile; display is not world state by itself.
ARC029	Effect	Persistent field state	Maintained support region with source, intensity, interactions, expiry and update schedule.
ARC030	Effect	Dispel and expiry	Removes or expires a declared effect through a typed inverse, counter or lifecycle rule.
ARC031	Conflict	Counterspell relation	Declared cancellation, redirection, reflection or suppression relation with priority and scope.
ARC032	Conflict	Converse and inverse declaration	Semantic converse/inverse is explicit; visual mirroring alone never creates it.
ARC033	Chance	Bounded randomness profile	Seeded finite draw, distribution and replay state; chance does not bypass guards or become hidden authority.
ARC034	Grammar	Ritual grammar	Bounded production grammar for multi-stage circles, chants, ingredients or coordinated casts.
ARC035	Projection	Gesture, sound and light view	Optional visual/audio/haptic projection of one spell graph; presentation can be replaced.
ARC036	Guard	Support guard	Caster, focus, target and support geometry/route are valid for the spell definition.
ARC037	Guard	Compatibility guard	World layer, target type, ward class, ownership and effect class are compatible.
ARC038	Guard	Cost and channel guard	Resources, cooldown, concentration slots and ritual dependencies are available.
ARC039	Guard	Authorization and consent guard	Player/role/target policy permits the proposed fictional effect in the selected experience.
ARC040	Guard	Invariant and hazard guard	World-law invariants, area budgets, persistence limits and forbidden combinations remain satisfied.
ARC041	Authority	Cast proposal	Input, AI, script or ritual creates a typed candidate with trace and reason codes only.
ARC042	Authority	Same-time spell resolution	Deterministic ordering, compatible merge, counter relation or explicit conflict event.
ARC043	Authority	Atomic effect patch	Applies resource, target, topology, knowledge and persistent-state changes together.
ARC044	Authority	Persistent update event	Ticks, transfers, expires or interrupts maintained effects through the same verifier.
ARC045	Certificate	Spell certificate	Records profile, definition hashes, ports, costs, guards, exactness, result and authority scope.
ARC046	Lineage	Arcane replay lineage	Stores cast/counter/expiry order, pre/post hashes, sources, branches and visible explanation.
ARC047	Query	Spell compiler and trace query	Resolves runes, types, dependencies, phases, costs, targets, counters and first failure.
ARC048	Adapter	Game, toy and performance adapter	RULEFORGE, SONORA, physical tiles or native/browser effects consume the same spell definition without changing authority.

Claims Ledger

Claim	Disposition	Reason
ARCANA implements real supernatural magic.	REJECT	It is a fictional rule-system cartridge for games, stories, learning and performance.
A drawn rune has intrinsic physical authority.	REJECT	Glyph geometry is a presentation face; semantics require a resolved OperatorCell and world-law profile.
Mirroring a rune automatically reverses a spell.	REJECT	Converse, inverse and counter relations must be declared explicitly.
A successful visual effect proves the cast committed.	REJECT	Presentation remains downstream of verified fictional world state.
One seed contains every future ritual or player choice.	REJECT	Seeds reconstruct only declared deterministic families; external novelty and commits remain logged.
Randomness may bypass cost or consent.	REJECT	Chance is a typed proposal input and all guards still apply.
Counterspells are universal inverses.	REJECT	Counter scope, priority, target class and residual persistent effects are profile data.
All fictional magic systems share the same mana or conservation law.	REJECT	Resources and invariants are world-profile specific.
Spell notation may silently change order.	REJECT	Dependencies, phases, grouping and effect order are explicit.
ARCANA is a safety-certified live-action effects system.	REJECT	No pyrotechnic, electrical, chemical or physical-effect authority is provided.
A spell certificate proves narrative truth outside its profile.	REJECT	Certificates prove only the declared fictional rules and replay result.
ARCANA replaces RULEFORGE, SONORA or graphics runtimes.	REJECT	It is a domain cartridge that uses those runtimes/adapters without absorbing their authority.

High-Impact Application Profiles

Profile	What the cartridge enables
Universal Magic-System Compiler	Ship multiple radically different magic worlds from data instead of hardcoding spells into separate engines.
Counterspell Arena	A deterministic PvP/PvE system where every ward, reflect, absorb, sever and residual effect has a visible reason trace.
Rune-Circuit Construction Toy	Combine VOLT-style ports, Definition Breaker tiles and tactile OperatorCells into a physical/digital spell laboratory.
Ritual Concert Engine	Let SONORA timing, harmony, voices and gestures drive cooperative spells whose state changes remain separate from the performance.
Arcane Immersive Theater	Synchronize projection, light, haptics and props from a safe fictional state ledger without claiming physical magic.
AI Spellsmith	Generate candidate spells, lore and visual notation while a small deterministic compiler rejects broken types, order, cost or world invariants.

Final Formal Definition

FORMAL DEFINITION

UGTOMS ARCANA 1.0 is a component-scoped fictional magic-system cartridge in which world laws, runes, typed spell ports, spell graphs, supports, targets, resources, cast schedules, ritual grammars, chance profiles, effects, persistent fields, counter-relations, cast proposals, atomic patches, certificates and replay lineage are explicit substrate data; visual, musical, tactile and physical manifestations remain projections; and only a verified cast under one declared world-law profile may alter fictional authoritative state.

Implementation sequence

- Freeze schemas for WorldLawProfile, RuneCell, SpellGraph, RitualGrammar, CounterProfile, SpellProposal and SpellCertificate.
- Implement the exact finite spell compiler, guard evaluator, resource reservation, conflict resolver and replay oracle.
- Ship Gatefold, Bind/Unbind/Sever, simultaneous counterspell and ritual-timing fixtures.
- Integrate RULEFORGE for game authority, SONORA for ritual timing/audio and a downstream browser/physical-rune projection.
- Add domain-specific balancing, accessibility, consent and live-performance safety only through separate expert-reviewed profiles.

RELEASE CONCLUSION

This 1.0 document freezes a domain-facing contract. It is deliberately useful before a production engine exists: implementations can now disagree visibly through typed profiles, reason codes and certificates instead of hiding differences behind the same symbol.